

RayaMart E-commerce Analysis

Project Focus: E-commerce Analysis

Project Kind: Data Analyst Project

Dataset Info:

Title: Messy E-commerce Dataset

Source: Kaggle

Link: <https://www.kaggle.com/datasets/saidnizam/messy-e-commerce-dataset?select=customers.csv>

No. of CSV content: 2

Dataset Problem:

1. Missing Values - strategic gaps across multiple columns
2. Duplicate Records - both exact and near-duplicate entries
3. Outliers - extreme values that will test your detection skills
4. Data Type Issues - numbers stored as strings, mixed encodings

Created Business Name: RayaMart

Business Question:

1. How does the customer demographic profile (age, gender, and subscription status) vary across different Indonesian provinces, and where should the marketing team allocate localized campaign budgets to address low engagement?
2. How long is the operational lag per province before a user makes their first transaction, indicating a need to optimize our regional onboarding funnel?
3. What is the total revenue and transaction completion rate across different payment methods (Credit Card, E-wallet, Bank Transfer, Cash) for each Indonesian province, and what common themes appear in the transaction review text for orders that were cancelled or left pending?

Data Question:

1. What is the demographic breakdown (age distribution and gender) of our customer base across different Indonesian provinces, and how does the subscription rate differ between these groups?
 - Needed Columns: age_group, gender, province, subscribe
2. What is the average operational timeline (lag) between a customer's signup date and their first transaction per province?
 - Needed Columns: regions, lag(transaction_date - signup_date)
3. What is the total revenue and transaction completion rate across different payment methods for each Indonesian province, and how do the common complaint themes in the customer reviews differ between cancelled and completed orders?
 - Needed Columns: province, payment_method, revenue, transaction_status, review_text

Analysis Steps:

Data Importation:

- Used Power Query to import the datasets into one

Total Number of Rows before cleaning:

- columns = 1010
- transactions = 8199

Data Cleaning:

Dataset: columns

- Used Power Query to:
 - Check the correct datatypes for each column
 - Delete Duplicates = 10 Rows
 - Confirmation through Research: Renamed the state column to province, as Indonesia does not have states
 - Replaced empty values in the province column with "Not Specified" as removing them can skew business metrics
 - Missing Ages: Used the Conditional Column to create bins for ages to deal with null values, where they are assigned as "Unknown". A bin size of 15 was created with the following ages following within either 15-29, 30-44, 45-59, and 60+.

Dataset: transactions

- Used Power Query to:
 - Confirmed if there are duplicates: there are no duplicates based on the transaction_id
 - Transformed the unit_price column into a currency type instead of a whole number
 - Replaced null values in discount_applied column to 0
 - Transformed discount_applied column into a percentage type

Total Number of Rows before cleaning:

- columns = 1000
- transactions = 8199

Feature Engineering:

Adding New Columns to transactions query

- Added a gross column by multiplying quantity and total_price
- Added a revenue column using Custom Column by using the following formula: $\text{revenue} = \text{gross} * (1 - \text{discount_applied})$
- Transformed the new column into a currency data type from a whole number

Creating a New Query

- Created a new query named rayamart by merging the customers and transactions query using a LEFT JOIN

- Selected relevant columns from transactions query: transaction_date, revenue, payment_method, transaction_status, review_text

Adding a New Column to the newly created query

- Used Custom Column to add lag_initial column by subtracting signup_date from transaction_date
 - Discovered 697 negative values in the new column which is not possible as the transaction_date should only be recorded after sign_up date
 - Decided to treat it as 0-Day Lag. It might have been a “Guest” purchase that triggered an instant account creation after checkout. This keeps the financial metrics 100% intact.
 - Created a new column called lag using Conditional Column in which if the lag_initial is less than 0 then the output will be 0 else it will copy the lag_initial values
 - Created a duplicate of review_text column and labeled it as temp_review in which all of the characters are in lowercase as Power Query is case-sensitive.
 - Used the temp_review table to create a Conditional Column called complaint_theme that contain the following themes and value under the themes:
 - Delayed Shipping: slow, delay, late
 - Damaged/ Defective Product: defect, damag, broken
 - Pricing Grievance: expens, overpric
 - Quality & Packaging Deficit: disappoint, poor, not neat
 - Expectation Gap: doesn't meet expectations, quality below expectations, doesn't match expectations, product doesn't expectations, product doesn't match picture, size doesn't match expectations, doesn't match my expectations
 - Brand Attrition: won't buy here again

EDA, Creating Pivot Table and Dashboard

Data Question #	Metric	Pivot Table Variables	Chart Type
1	Total No. of customers per province and by subscription	Rows: province; Values: count of customer_id; Slicer/Filter: subscribe (used together with Gender Distribution Per Province for optimal filtering)	Clustered Bar Chart: used as provinces have long names, and to easily visualize which province has more customers
	Gender Distribution per Province and by subscription	Columns: gender; Rows: province; Values: % of grand total of count of customer_id; Slicer/Filter: subscribe	100% Stacked Bar Chart: better for visualizing two sub-categories
2	Average	Rows: province; Values:	Horizontal Bar Chart: used as province names

	Operation Timeline (Lag) per Province	Average lag	are long and better for identifying which province has the highest Lag
3	Revenue by Payment Method and Transaction Status	Columns: payment_method; Rows: province; Values: sum of revenue; Slicer: transaction_status, payment_method	Clustered Bar Chart: easier to spot which payment method is mostly used per province
	Transaction Rate by Transaction Status	Axis: Province; Legend: payment_method; Values: count of transaction_status; Slicer: transaction status, payment_method	Horizontal Bar Chart: easier to visualize which province has the highest transaction rate, filtered by transaction status
	Total Number of Complaint Themes by Transaction Status	Columns: transaction_status; Rows: complaint_theme; Values: Count of complaint_theme	Horizontal Bar Chart: appropriate for visualizing side-by-side each complaint theme and their total number filtered by transaction status

Answering the Business Questions:

- How does the customer demographic profile (age, gender, and subscription status) vary across different Indonesian provinces, and where should the marketing team allocate localized campaign budgets to address low engagement?

Most of the buyers have not specified where they come from. As a result, the “not specified” group has the largest population recorded.

Age:

- Most buyers are in the 30-44 and 45-49 age groups. The ‘Not specified’ group, Kepaluan Riau, and Sulawesi Utara have the most buyers in the mentioned age group.
- The lowest number of buyers by age group would be those in the ‘Unknown’ group and 60+. Specifically, the marketing can target the provinces of Sumatera Selatan, Bali, and Jawa Timur and others who are on the lower end for their campaign.

Gender:

- Most of the buyers are female. The ‘Not specified’ group, Banten, and Riau have the highest female customers.
- The marketing team can target males for their campaign, and specifically provinces such as Sulawesi Barat, Sulawesi Selatan, Nusa Tenggara Timur, and other provinces who are on the lower end as well.

Subscription Status:

- The ‘Not specified’ group, Banten, and Nusa Tenggara Barat have the highest subscription.
- While for those who have not subscribed mostly came from Papua, ‘Not Specified’ group, and Sumatera Utara. The marketing can target these provinces for their campaign as well as other groups.

2. How long is the operational lag per province before a user makes their first transaction, indicating a need to optimize our regional onboarding funnel?

The provinces of Aceh, Sulawesi Barat, and Jambi have the highest Average Operation Timeline (Lag). The marketing team can target their campaign among these provinces and others that have high Lag.

3. What is the total revenue and transaction completion rate across different payment methods (Credit Card, E-wallet, Bank Transfer, Cash) for each Indonesian province, and what common themes appear in the transaction review text for orders that were cancelled or left pending?

Total Revenue (Transaction Status = Completed):

- The 'Not specified' group, Banten, and Nusa Tenggara Barat have the highest completed revenue. For both 'Not specified' and Banten, the cash payment method has the highest transaction. For Nusa Tenggara Barat, the bank transfer method has the highest transaction.
- Sulawesi Barat, Sulawesi Tengah, and Jambi have the lowest completed revenue. For Sulawesi Barat, the credit card has the highest transaction. For Sulawesi Tengah, the use of cash has the highest transaction. For Jambi, the use of e-wallet has the highest transaction.

Transaction Completion Rate (Transaction Status = Completed):

- 'Not specified' has the highest transaction completion rate. Can be drilled down using filters for more specific metrics.

Transaction Complaint Themes:

- For all transaction status types (Cancelled, Completed, Pending, Refunded), they have the same top complaint themes. In order, the most complained of is the Quality & Packaging Deficit, followed by Delayed Shipping and Pricing Grievance.